

DL Midsem Notes — Section 10

Convolutional Neural Networks

1 Theory

1.1 The Convolution Operation

A filter (kernel) of size $F \times F$ slides over the input, at each position computing a dot product between the filter weights and the patch of input it currently covers, producing one output value (a **feature map**). Parameters that control the output size:

- **Stride** S : how many pixels the filter moves per step.
- **Padding** P : how many pixels of (usually zero) border are added around the input before convolving — used to control output size, most commonly to keep it equal to the input size ("same" padding).

1.2 Output Dimension Formula

For an input of size $W_1 \times H_1$, filter size F , stride S , padding P :

$$W_2 = \frac{W_1 - F + 2P}{S} + 1, \quad H_2 = \frac{H_1 - F + 2P}{S} + 1$$

(derivation in Q1). "Same" padding (output size = input size, stride 1) requires $P = (F - 1)/2$.

1.3 Parameter Count

With K filters, each of size $F \times F$ and depth d (matching the input's channel depth), plus one bias per filter:

$$\text{\#params} = K(F \cdot F \cdot d + 1)$$

Note this does *not* depend on the spatial size of the input at all — weight sharing (the same filter slides over the whole image) is exactly what keeps CNN parameter counts small compared to a fully-connected layer over the same input.

1.4 Pooling

Max (or average) pooling reduces spatial size by summarizing each small window (e.g. 2×2) to one value — reduces computation, adds a degree of translation invariance (a small shift in the input barely changes the max/average of a window), and has no learnable parameters.

1.5 Filter Size Extremes

$F = 1$ (a 1×1 convolution): the output has the same spatial size as the input; it acts purely as a per-pixel linear recombination across channels (no spatial mixing at all). $F = n$ (filter as large as the entire image): with no padding the output collapses to a single 1×1 value per filter — effectively a fully-connected layer over the whole input, all spatial information immediately compressed into single numbers.

2 Practice Questions

Q1 (PYQ-style, CS6510-3 Q1a / CS6510-4 Q1A — asked twice, high priority). Given a filter of width F and depth d , applied to an image of dimension (W_1, H_1) , derive the equation for (W_2, H_2) assuming S strides and P padding. If there are K kernels, what is the number of parameters formed?

Q2 (PYQ-style, CS6510-3 Q3b, reworded as direct question). An input image I of size $n \times n$ is convolved once with a filter of size 1×1 , and separately with a filter of size $n \times n$ (stride 1, no padding, in both cases). What happens to the output in each case, and why?

Q3 (PYQ-style, CS6510-5 Q5a). For the architecture $\text{INPUT}(128 \times 128 \times 3) \rightarrow \text{CONV-9-32} \rightarrow \text{POOL-2} \rightarrow \text{CONV-5-64} \rightarrow \text{POOL-2} \rightarrow \text{CONV-5-64} \rightarrow \text{POOL-2} \rightarrow \text{FC-10} \rightarrow \text{FC-3}$ (CONV- F - K = filter size $F \times F$, K filters, stride 1, no padding; POOL-2 = 2×2 , stride 2), compute the feature map size and parameter count at each layer.

Q4 (PYQ-style, CS6510-5 Q6a). A 5×5 input

$$\begin{pmatrix} 1 & 2 & 0 & 1 & 2 \\ 0 & 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 2 & 0 \\ 2 & 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 & 2 \end{pmatrix}$$

is convolved (stride 1, no padding) with the 3×3 filter $\begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}$. Find the convolution output.

Then apply 2×2 max pooling with stride 1 to that output. Finally, what padding size would make the *convolution* output the same size as the original 5×5 input?

3 Answers

A1. Padding adds P pixels on each side, so the effective input size is $W_1 + 2P$. A filter of size F sliding with stride S fits $\lfloor (W_1 + 2P - F)/S \rfloor + 1$ times along that axis (each step after the first consumes S more pixels of the effective width until the filter no longer fits):

$$W_2 = \frac{W_1 - F + 2P}{S} + 1, \quad H_2 = \frac{H_1 - F + 2P}{S} + 1$$

Each of the K filters has $F \times F \times d$ weights plus 1 bias, and filters don't share parameters with each other:

$$\# \text{params} = K(F^2 d + 1)$$

A2. With $F = 1$: using the output formula with $S = 1, P = 0$: $W_2 = (n - 1 + 0)/1 + 1 = n$ — output is the same $n \times n$ size as the input; each output pixel is just a linear recombination of the input channels at that exact pixel location, with zero spatial mixing (no neighboring pixels are involved at all). With $F = n$: $W_2 = (n - n + 0)/1 + 1 = 1$ — the filter covers the entire image at once, so there's exactly one valid position; the output collapses to a single 1×1 value per filter, i.e. the convolution degenerates into a fully-connected layer over the whole image, and all spatial/local structure is discarded in one step.

A3. Using $W_2 = (W_1 - F)/S + 1$ with $S = 1, P = 0$ for conv layers, and halving for each 2×2 /stride-2 pool:

Layer	Output size	Parameters
INPUT	$128 \times 128 \times 3$	—
CONV-9-32	$(128 - 9 + 1) = 120 \rightarrow 120 \times 120 \times 32$	$32(9^2 \cdot 3 + 1) = 32(244) = 7808$
POOL-2	$60 \times 60 \times 32$	0
CONV-5-64	$(60 - 5 + 1) = 56 \rightarrow 56 \times 56 \times 64$	$64(5^2 \cdot 32 + 1) = 64(801) = 51264$
POOL-2	$28 \times 28 \times 64$	0
CONV-5-64	$(28 - 5 + 1) = 24 \rightarrow 24 \times 24 \times 64$	$64(5^2 \cdot 64 + 1) = 64(1601) = 102464$
POOL-2	$12 \times 12 \times 64$	0
FC-10	flatten $12 \cdot 12 \cdot 64 = 9216 \rightarrow 10$	$9216(10) + 10 = 92170$
FC-3	$10 \rightarrow 3$	$10(3) + 3 = 33$

Total parameters = $7808 + 51264 + 102464 + 92170 + 33 = 253,739$.

A4. Sliding the 3×3 filter over the 5×5 input (stride 1, no padding) gives a 3×3 output (using $W_2 = (5 - 3)/1 + 1 = 3$). Computing each dot product:

$$\text{Conv output} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

Applying 2×2 max pooling with stride 1 (output size $(3 - 2)/1 + 1 = 2$), taking the max of each 2×2 window:

$$\text{top-left: } \max(-1, 0, 0, -1) = 0, \quad \text{top-right: } \max(0, 0, -1, 1) = 1$$

$$\text{bottom-left: } \max(0, -1, 1, 0) = 1, \quad \text{bottom-right: } \max(-1, 1, 0, -1) = 1$$

$$\text{Max-pooled output} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

To keep the convolution output the same size as the 5×5 input (i.e. "same" padding, stride 1): $P = (F - 1)/2 = (3 - 1)/2 = \boxed{1}$.